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(54) **VENT COVER ASSEMBLIES FOR RAILROAD HOPPER CARS**

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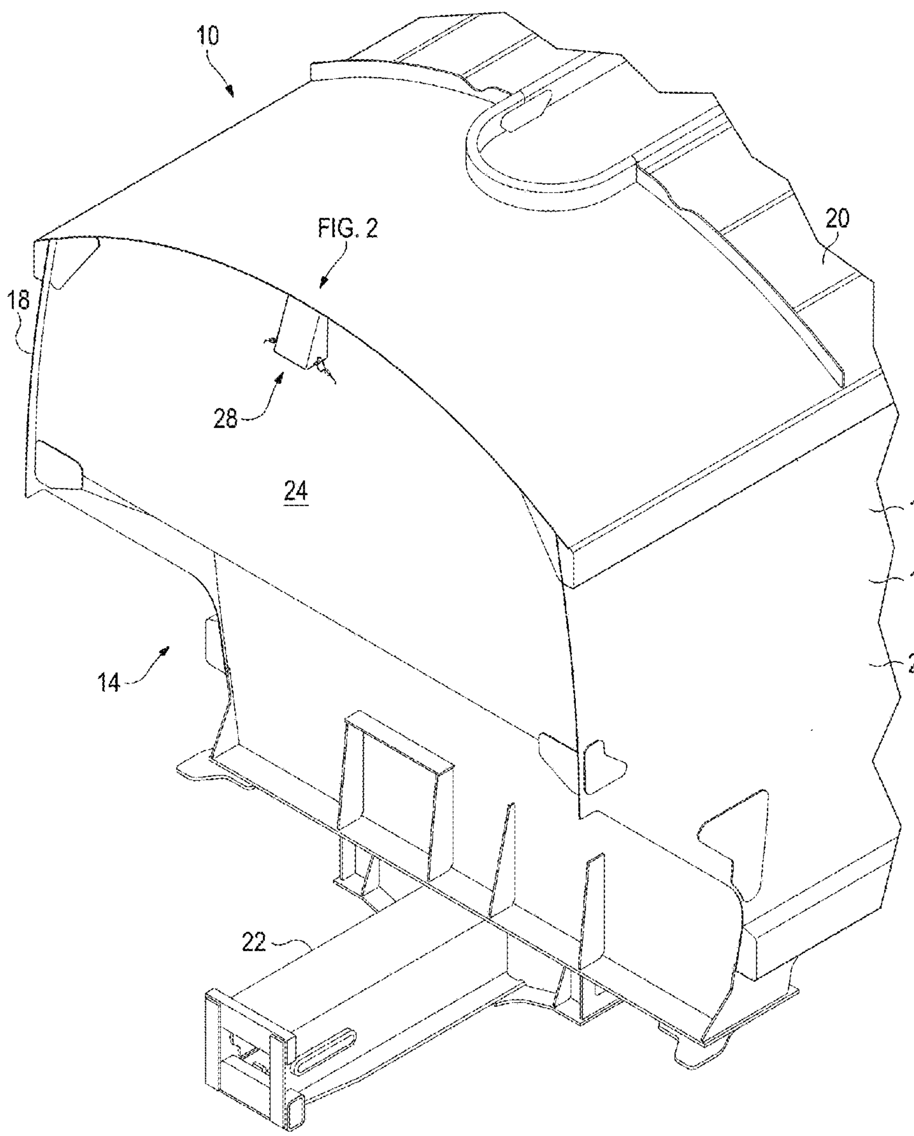
CPC ***B61D 27/009*** (2013.01)

(57) **ABSTRACT**

Vent cover assemblies for a railway car are shown and disclosed. In one embodiment, the vent cover assembly includes a base housing having opposed first and second end portions. The second end portion includes opposed first and second walls and an opening therebetween. The first wall includes a first slot and the second wall includes a second slot. The vent cover assembly additionally includes a cover that spans the opening and includes opposed first and second tabs. The first tab is sized to be received in the first slot and the second tab is sized to be received in the second slot.



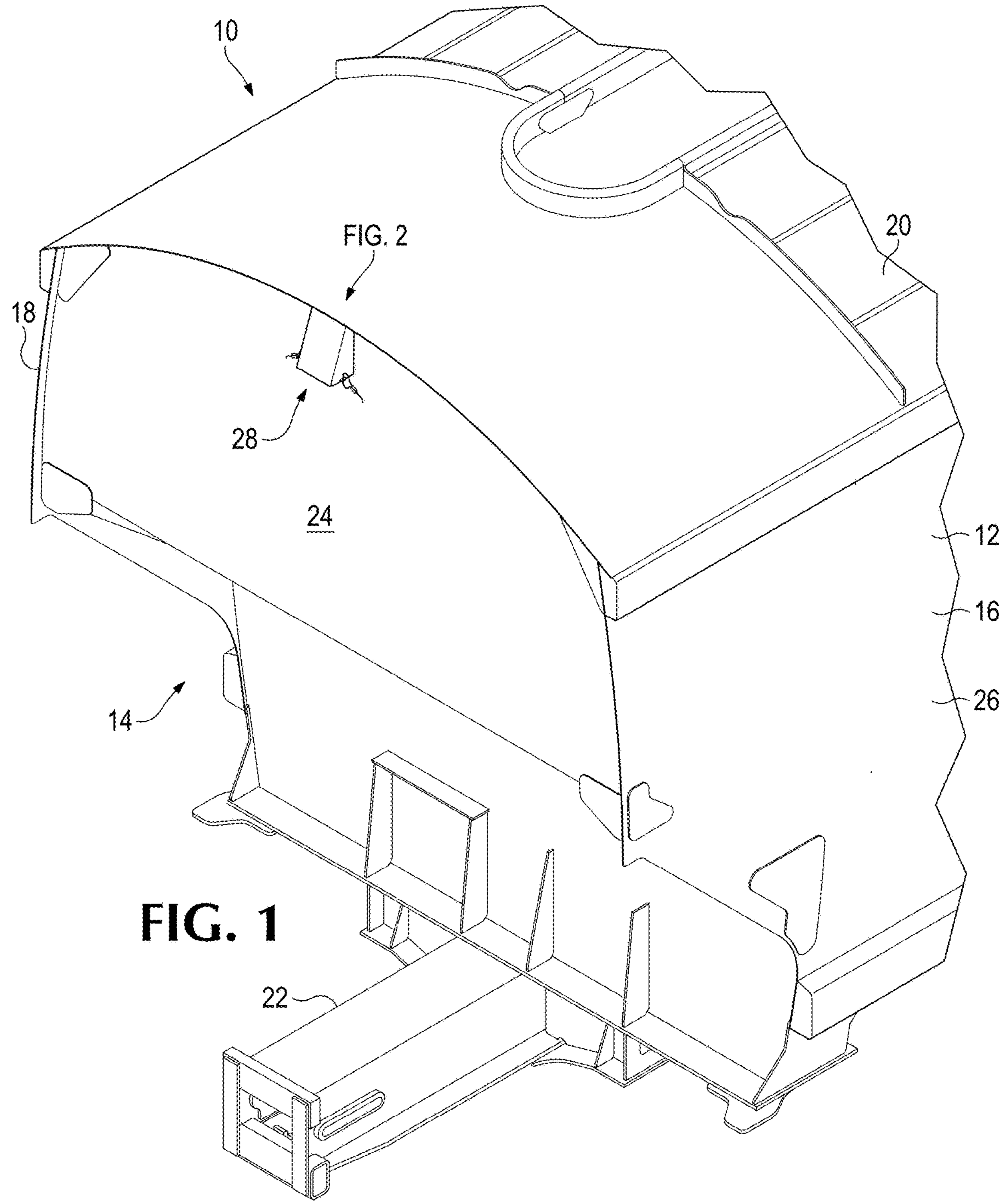


FIG. 1

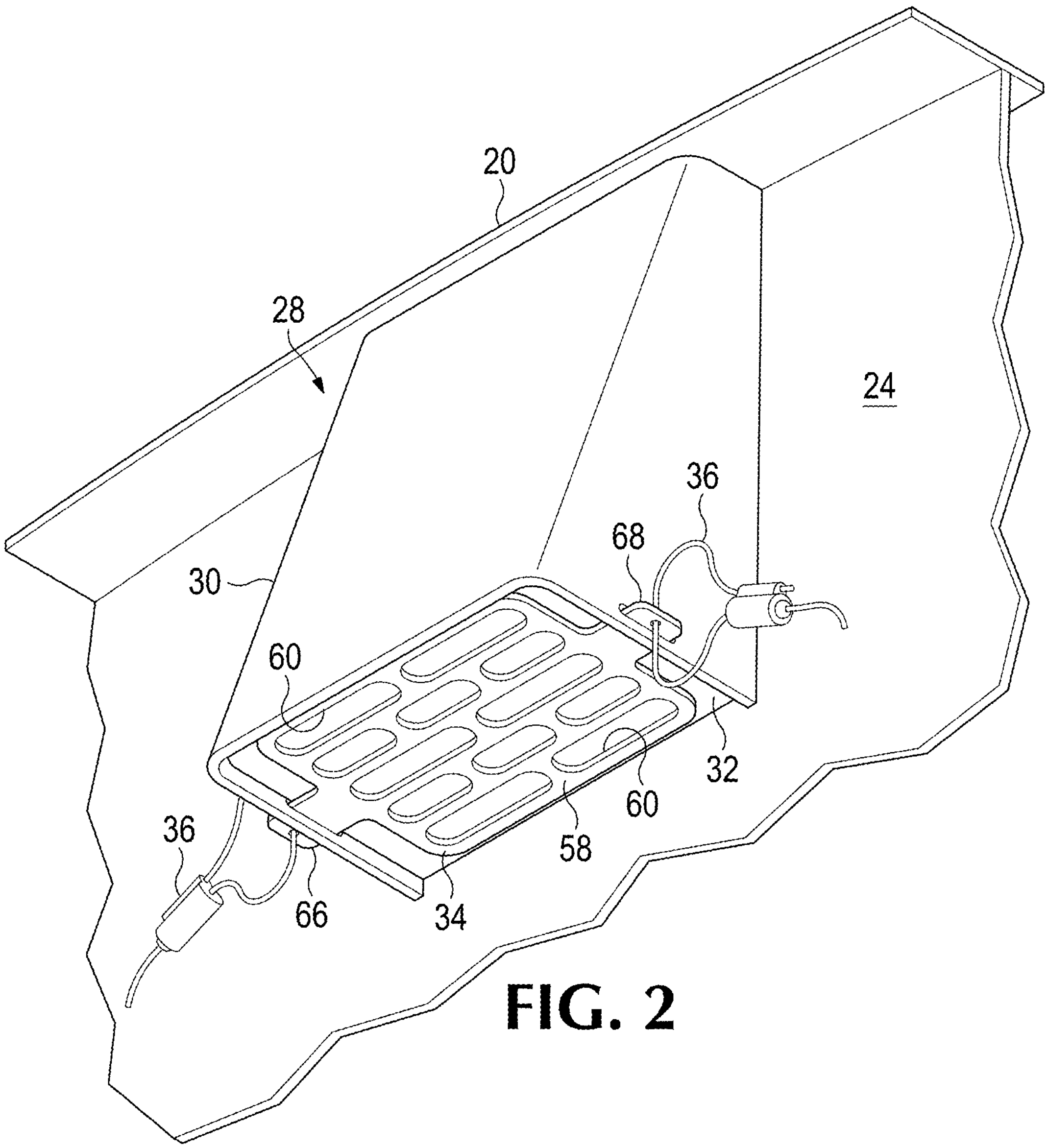


FIG. 2

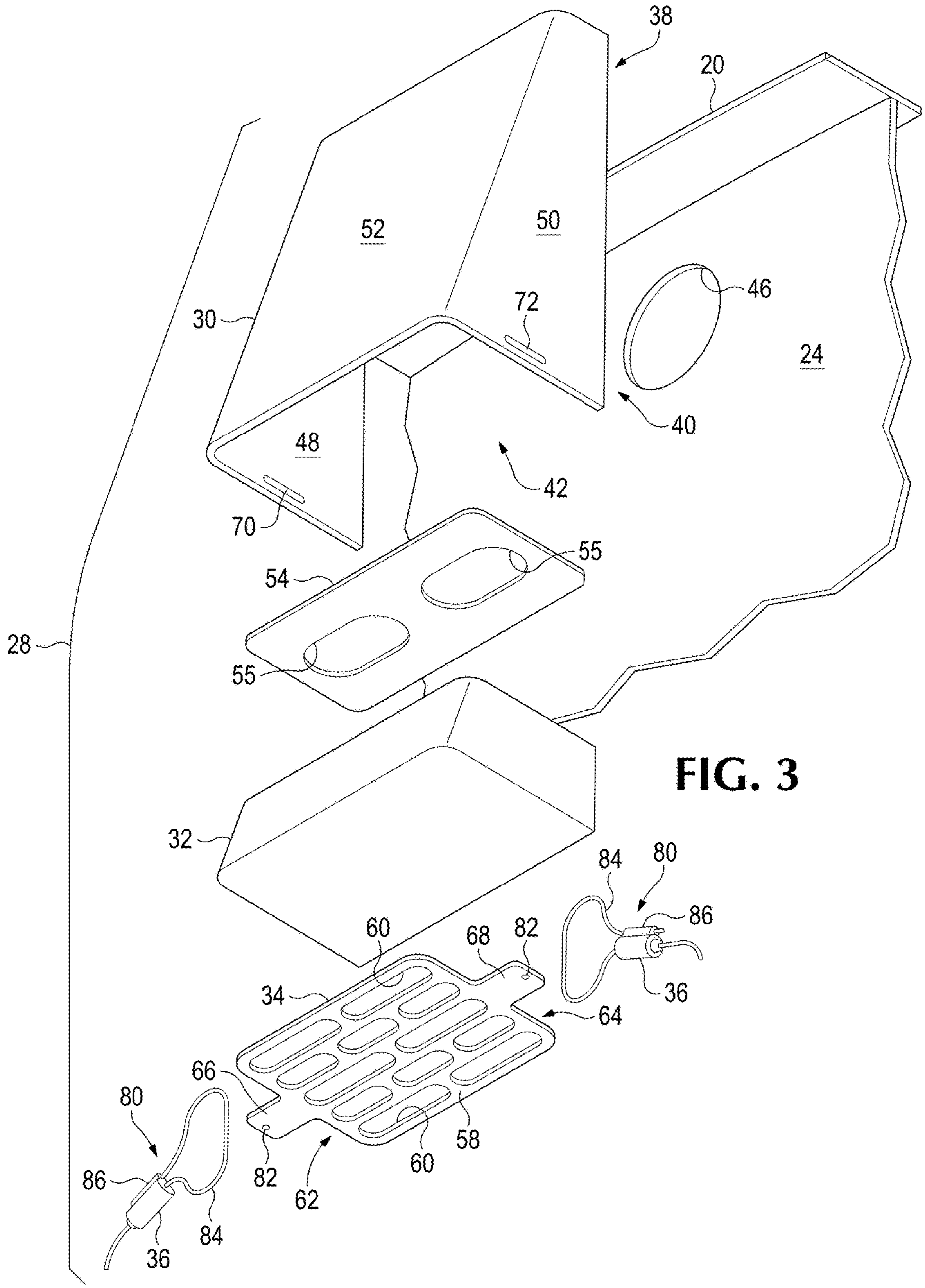

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FIG. 3

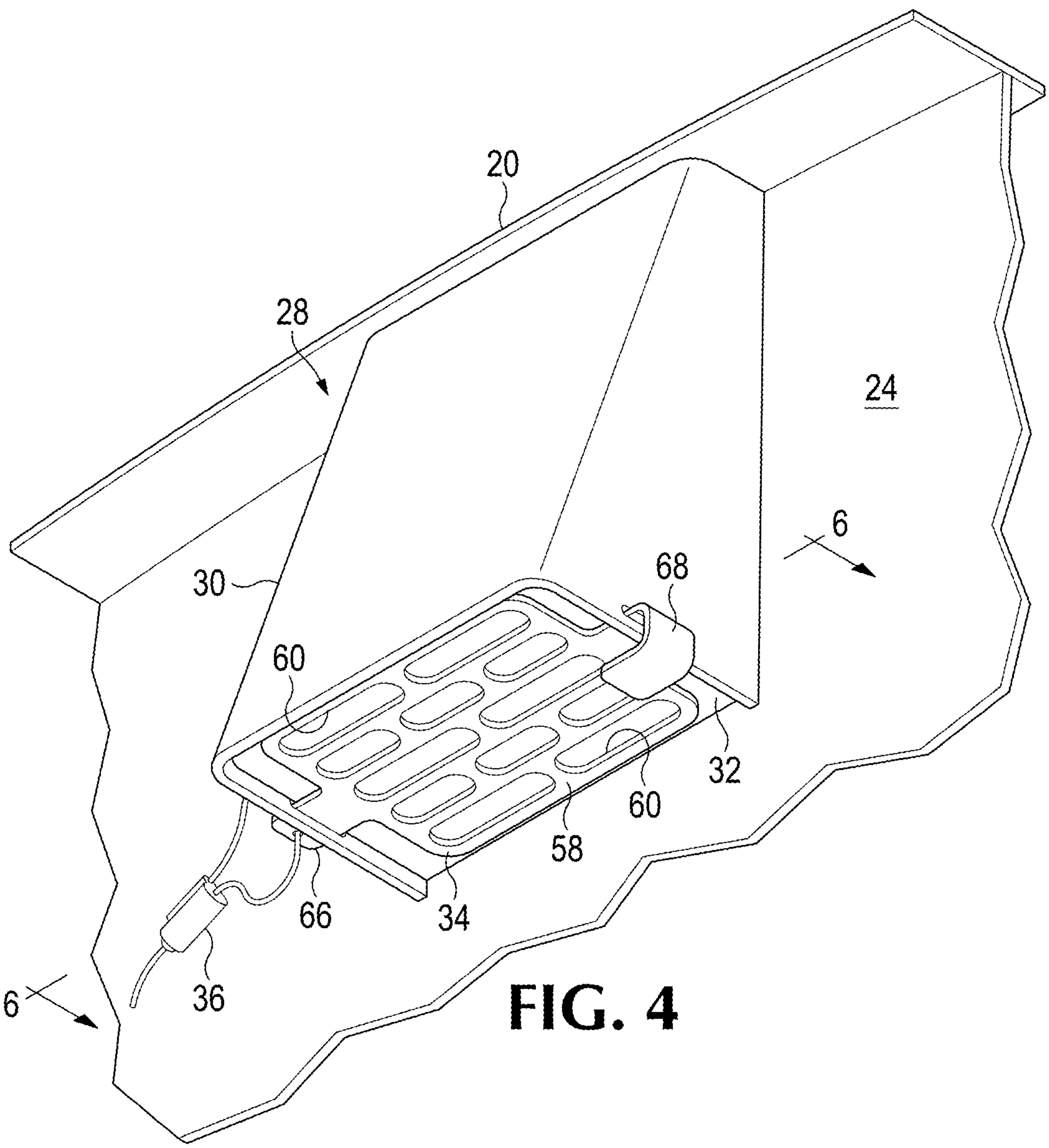

FIG. 4

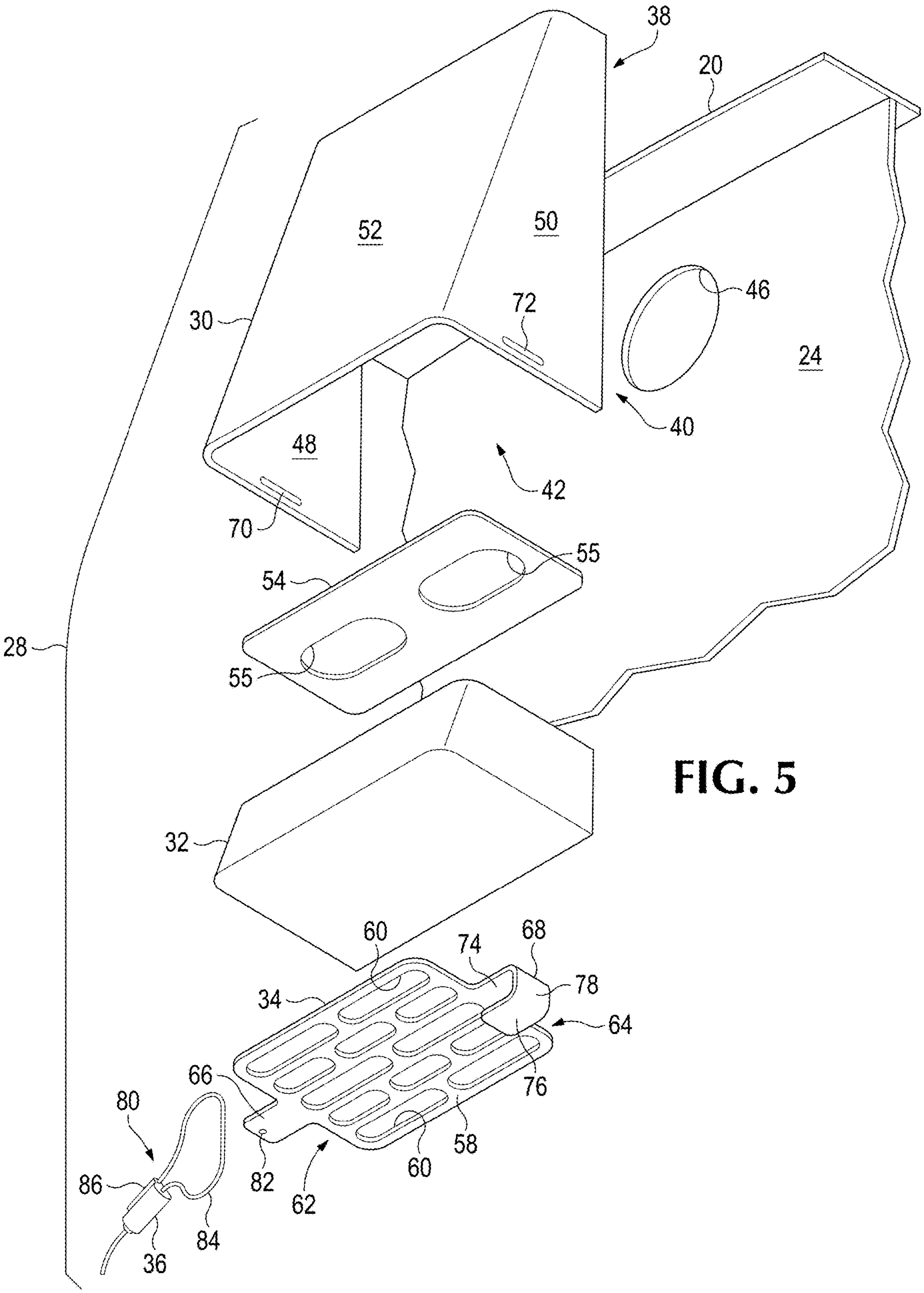
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FIG. 5

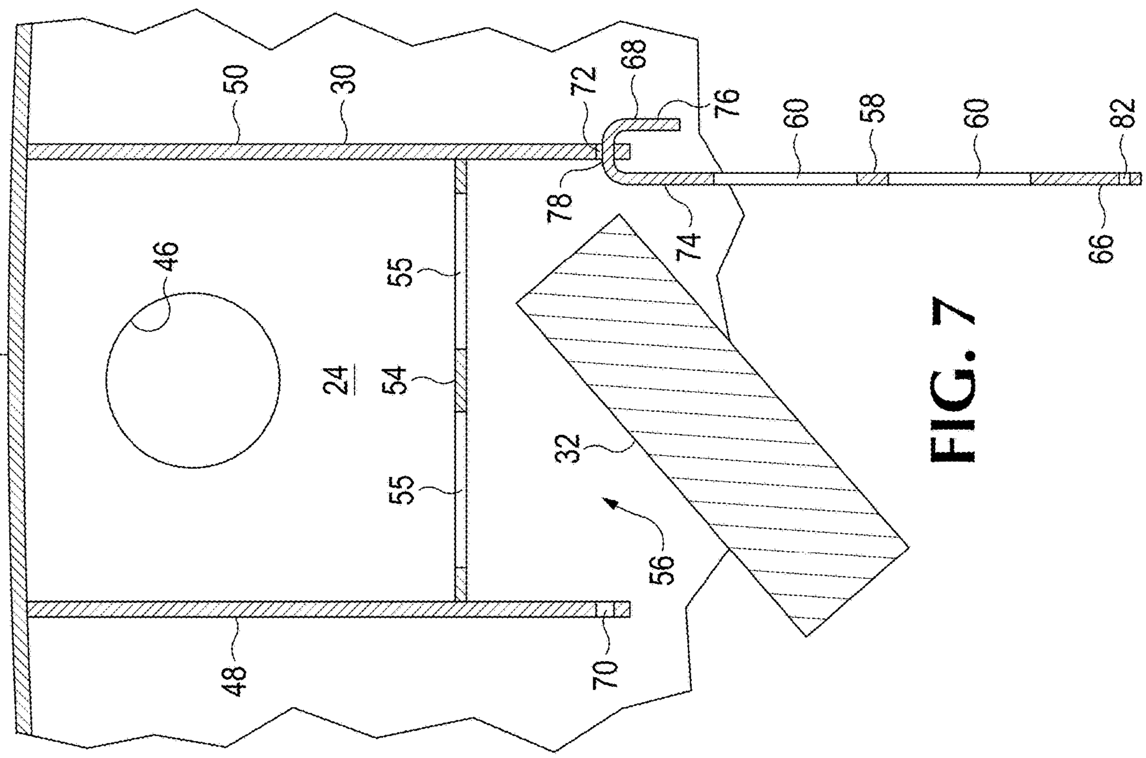
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FIG. 7

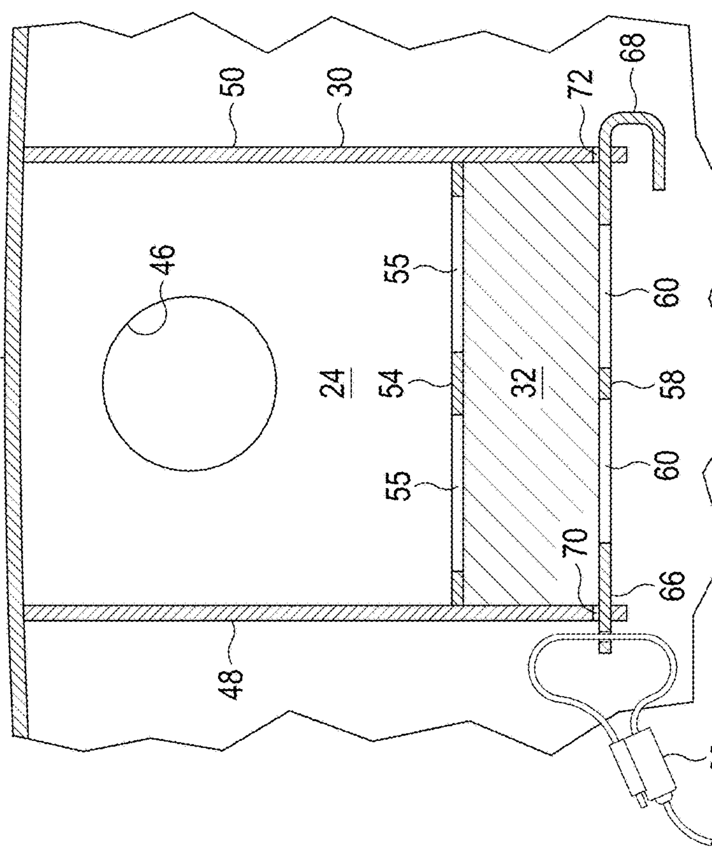
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FIG. 6

VENT COVER ASSEMBLIES FOR RAILROAD HOPPER CARS

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/563,182 filed on Mar. 8, 2024 and entitled "Vent Cover Assemblies for Hopper Cars." The complete disclosure of the above application is hereby incorporated by reference for all purposes.

BACKGROUND

[0002] Railroad cars, such as hopper cars, generally include vents and vent cover assemblies. The purpose of the vent cover assembly is to prevent contamination of the product inside the railcar from outside weather elements, insects, or people while allowing air to flow freely between the inside of the railcar and the atmosphere. Previous designs included vent cover assemblies having filters that were prone to falling out of the housing and that were not tamperproof.

[0003] What is desired, therefore, is a vent cover assembly having a filter that is easily installed and replaced, and provides adequate protection from the elements, a secure installment, and clear evidence in the event of tampering.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] For a better understanding of the invention, and to show how the same may be carried into effect, reference will now be made, by way of example to the accompanying drawings, which:

[0005] FIG. **1** is a partial view of a railroad hopper car shown without its wheeled trucks;

[0006] FIG. **2** is a partial view of an example of a vent cover assembly of the railroad hopper car of FIG. **1**;

[0007] FIG. **3** is an exploded view of the vent cover assembly of FIG. **2**;

[0008] FIG. **4** is a partial view of another example of a vent cover assembly of the railroad hopper car of FIG. **1**;

[0009] FIG. **5** is an exploded view of the vent cover assembly FIG. **4**; and

[0010] FIGS. **6-7** are sectional views of the vent cover assembly of FIG. **4** taken along **6**—**6** in FIG. **4**, showing an example of opening a cover of the vent cover assembly to remove filter media.

DETAILED DESCRIPTION

[0011] Referring to FIG. **1**, a railroad hopper car **10** is shown. The railroad hopper car includes a car body **12** carried on a pair of wheeled trucks (not shown) and having a pair of opposed ends with one end **14** shown in FIG. **1**. Additionally, railroad hopper car **10** includes a pair of opposite sides **16** and **18** and a roof **20** attached to and supported by the ends and sides. A center sill **22** may extend through the entire length of the car body. Moreover, the railroad hopper car includes one or more cargo hoppers. Furthermore, railroad hopper car **10** includes opposed end walls and opposed side walls. One end wall **24** and a portion of one side wall **26** are shown in FIG. **1**. One or both end walls of the railroad hopper car may include a vent assembly **28**. The railroad hopper car may additionally, or alternatively, include one or more components, such as shown in U.S. Pat. Nos. 10,807,615 and 12,122,431. The complete disclosures of the above patents are hereby incorporated by reference for all purposes.

[0012] Referring to FIGS. **2-3**, an example of vent assembly **28** is shown. The vent assembly includes a base housing **30**, a filter or filter media **32**, a cover **34**, and one or more securing structures **36**. The base housing includes opposed first and second end portions **38**, **40**. Second end portion **40** includes a base opening **42**. The first end portion of the base housing is attached (e.g., welded) adjacent to and/or in fluid communication with at least one vent opening or hole that open up to the hopper(s) or main compartment of the hopper car, such as a vent opening **46** in end wall **24** of railroad hopper car **10**. Base housing **30** may be any suitable shape (s). In the example shown in FIGS. **2-3**, base housing, the base housing forms a hollow truncated pyramid or a frustum of a pyramid with the end wall. Alternatively, the base housing may form a hollow rectangular prism, a hollow cube, and/or other suitable shape(s) with the end wall. The base housing in FIGS. **2-3** includes opposed, spaced, and/or parallel first and second side walls **48**, **50** and a base wall **52** disposed between and/or perpendicular to the first and second side walls. The side walls and base walls may be attached to or formed with each other.

[0013] In some examples, base housing **30** may include one or more internal walls. In the example shown in FIGS. **2-3**, base housing includes a middle or internal wall **54** that is attached to, or formed with, first and second side walls **48**, **50** and base wall **52** to subdivide the interior of the base housing. The internal wall may be perpendicular to the first and second side walls and the base wall, as shown in FIGS. **2-3**. The internal wall includes a plurality of openings **55** to allow vented gases to flow through that wall. Those openings may be any suitable shape, such as circular, oblong or, as shown in FIGS. **2-3**, openings **55** may be stadium-shaped and may include the same and/or different lengths. The first and second side walls, the base wall and the internal wall collectively form a pan or receptacle **56** (best shown in FIG. **7**) sized to contain or receive the filter media and/or cover of the vent assembly.

[0014] Filter **32** may include any suitable structure to prevent contamination of the product contained within the hopper from outside weather elements, insects, or people, while allowing air to flow freely between the inside of the hopper and the atmosphere. For example, the filter may be an open-celled foam filter sized and/or shaped to fit within receptacle **56**.

[0015] Cover **34** may include any suitable structure to secure the filter to the base housing and/or prevent the filter from falling out of the housing. The cover includes a planar base **58** having a plurality of holes or openings **60**. Those holes may be any suitable shape(s), such as circular, oblong and/or, as shown in FIGS. **2-3**, openings **54** may be stadium-shaped. The planar base includes first and second opposed end portions **62**, **64**. A first tab or end tab **66** is attached to, or formed with, first end portion **62**, and a second tab or end tab **68** is attached to, or formed with, second end potion **64**. The first and second end tabs are sized to be received in and/or extend through slots in the base housing. In the example shown in FIGS. **2-3**, first end tab **66** is sized to be received in a first slot **70** on first side wall **48** and second end tab **68** is sized to be received in a second slot **72** on second side wall **50**. The end tabs may be any suitable length(s) and/or shape(s). In the example shown in FIGS. **2-3**, the end tabs are planar and sized to extend through the opposed slots such that the holes on the end tabs are external the housing and may be used with one or more securing structures **36**. The cover is thus removable as compared to fixed bottom cover designs, which allows for easy installation and replacement of the filter.

[0016] In another example, first end tab **66** may be similar or identical to the example shown in FIGS. **2-3** and second end tab **68** may be sized longer than the first end tab and/or be shaped to curve around the housing, as shown in FIGS. **4-5**. For example, the end tab may be C-shaped (curvilinear and/or rectilinear) and/or have two parallel portions **74**, **76** with an intermediate portion **78** that is disposed between and connects the parallel portions and/or that is perpendicular to the parallel portions. The example of FIGS. **4-5** allows for the bottom cover to remain connected and/or supported by the housing while the filter is being replaced to prevent the bottom cover from falling or being misplaced, as shown in FIGS. **6-7**. The cover as shown in FIG. **6** may be referred to be in a closed position in which the first tab is received in the first slot and the second tab is received in the second slot. Additionally, the cover as shown in FIG. **7** may be referred to be in an open position in which the first tab is not received in either of the first or second slots (and spaced from both slots) and the second tab (or its intermediate portion) is received in the second slot. Because the second tab remains in the second slot in the open position, the cover remains connected to the rest of the vent cover assembly. The filter is enclosed and/or contained within the receptacle when the cover is in the closed position, and is removable from the receptacle when the cover is in the open position.

[0017] Securing structure(s) **36** secure cover **34** (and consequently filter **32**) to base housing **30** and allows an operator to visually determine whether someone has disturbed or tampered the securing structure(s). In the examples shown in FIGS. **2-5**, the securing structures are in the form of one or more cable seals **80** that are inserted into holes **82** of one or more of the end tabs of the cover. Each of the cable seals includes a cable **84** sized to be received in the holes of the ends tabs and a locking mechanism **86** that secures the cable to the locking mechanism. Although two cable seals are shown in FIGS. **2-3**, a single cable seal that is sized to be received in both holes of the end tabs (or in one hole when the other end tab does not include a hole as shown in FIGS. **4-5**) may alternatively be used. Each of the cable seals may have a unique alphanumeric code that can be recorded and/or traced. In the event of tampering, the operator would be able to tell by either a damaged seal or a cable seal having an alphanumeric codes that does not match what was previously recorded. An operator may cut or unlock the cable seals to remove the bottom cover and replace the filter. Other securing structures **36** may be used to secure the cover to the base housing, such as zip ties, padlocks, cable padlocks, combination padlocks, etc.

[0018] The above vent cover assembly allows for easy installation and replacement of the filter and provides adequate protection from the elements, a secure installment, and clear evidence in the event of tampering.

[0019] It will be appreciated that the invention is not restricted to the particular embodiment that has been described, and that variations may be made therein without departing from the scope of the invention as defined in the appending claims, as interpreted in accordance with principles of prevailing law, including the doctrine of equivalents or any other principle that enlarges the enforceable scope of a claim beyond its literal scope. Unless the context indicates otherwise, a reference in a claim to the number of instances of an element, be it a reference to one instance or more than one instance, requires at least the stated number of instances of the element but is not intended to exclude from the scope of the claim a structure or method having more instances of that element than stated. The word "comprise" or a derivative thereof, when used in a claim, is used in a nonexclusive sense that is not intended to exclude the presence of other elements or steps in a claimed structure or method.

What is claimed is:

1. A vent cover assembly for a railway car, comprising:

a base housing having opposed first and second end portions, the second end portion having opposed first and second walls and an opening therebetween, the first wall having a first slot and the second wall having a second slot; and

a cover that spans the opening and includes opposed first and second tabs, the first tab is sized to be received in the first slot and the second tab is sized to be received in the second slot.

2. The vent cover assembly of claim **1**, further comprising filter media disposed within the base housing.

3. The vent cover assembly of claim **2**, wherein the filter media is an open-celled foam filter.

4. The vent cover assembly of claim **1**, wherein the base housing includes first and second opposed walls and a base wall disposed between the first and second walls.

5. The vent cover assembly of claim **4**, wherein the first wall includes the first slot and the second wall includes the second slot.

6. The vent cover assembly of claim **1**, wherein the cover includes an elongate cover base having opposed first and second end portions, and wherein the first tab is attached to, or formed with, the first end portion, and the second tab is attached to, or formed with, the second end portion.

7. The vent cover assembly of claim **6**, wherein the elongate cover base includes a plurality of openings.

8. The vent cover assembly of claim **7**, wherein one or more of the plurality of openings is stadium-shaped.

9. The vent cover assembly of claim **1**, wherein at least one of the first or second tabs includes an aperture.

10. The vent cover assembly of claim **9**, further comprising a cable seal having a cable and a locking mechanism, the aperture of the at least one of the first or second tabs being sized to receive the cable.

11. The vent cover assembly of claim **9**, wherein each of the first and the second tabs includes the aperture.

12. The vent cover assembly of claim **1**, wherein one of the first or second tabs is C-shaped.

13. The vent cover assembly of claim **12**, wherein one of the first or second tabs includes parallel and opposed first and second portions and a third portion disposed between and perpendicular to the first and second portions.

14. The vent cover assembly of claim **13**, wherein the first, second, and third portions are sized to be received in the first or second slots.

15. The vent cover assembly of claim **14**, wherein the cover is movable between a closed position in which the first tab is received in the first slot and the second tab is received in the second slot, and an open position in which the first tab is spaced from the first and second slots and the third portion of the second tab is received in the second slot.

16. The vent cover assembly of claim **1**, further comprising a middle wall attached to, or formed with, one or more internal walls of the base housing.

17. The vent cover assembly of claim **16**, wherein the base housing and the middle wall define a filter area sized to receive filter media.

18. A railcar having an end wall with a vent opening, wherein the first end portion of the vent cover assembly of claim **1** is attached to, or formed with, the end wall adjacent to the vent opening such that the base housing is in fluid communication with the vent opening.

19. The railcar of claim **18**, wherein the railcar is a railroad hopper car.

* * * * *